ENS3553/ENS6160 Signals and Systems

Laboratory Experiment 3

Sampling, Windowing and Spectral Analysis of Signals

Prerequisites:

- A thorough review of the theory on Windowing and Sampling as well as Discrete Fourier Transforms presented in Modules 10 and 11

Objectives:

- The aim of this laboratory is to use MATLAB and the oscilloscope to see the effects of sampling and aliasing, and windowing
- Additionally, using MATLAB to perform the Fast Fourier Transform (FFT).

Post-Lab

- Make sure you work through Lecture 10 on Sampling Theory prior to attempting the post-lab exercises and refer to the MATLAB documentation as required
- Show your complete working for the following exercises in your lab results sheet:

1. What is the Nyquist frequency for a signal?
2. What are the resulting frequencies for the following input sinusoids 500Hz, 2.5KHz, 5KHz and 5.5KHz if the signals are sampled with a sampling frequency of 5KHz?
3. Sketch the time domain waveform of a unit rectangular pulse of width *T* and its frequency domain spectrum. Clearly list all key points on the sketches.
   Hint: Refer to Module 7.
4. Sketch the time domain waveform and frequency spectrum of a Hanning window. (*Note:* The frequency response is not in unit materials – you will have to find it)
5. Give an explanation of what the MATLAB command *fft* does and how it is used. *Hint:* You can use the Matlab *help* function.
6. Download the Matlab file *prelab3.m* attached along with this instruction sheet. Open the file in the editor and understand what the script is doing. Run the script, print out the plot and paste a copy into your lab result sheet.
   a. Explain the shape of the spectral plots. Why are they not the theoretical 'impulses' you would expect for a sinusoid?
   b. Explain the effect of changing the value of N. *Hint:* Refer Module 10.
7. Extend the sample by changing the first line to `n = [0:59];`
   Run the script again and paste a copy of the plot into your lab result sheet. Explain the difference in shape and height .

Laboratory Experiment

Equipment:

- PC with MATLAB installed
- Function generator (33220A)
- Digital Storage Oscilloscope (DSO-X 3024A)

Procedure:

A. Sampling of a signal

1. Download the *cosgen2* function from Blackboard and save to the Desktop. Make sure that the name is still *cosgen2.m*. Open MATLAB and change the working directory to 'Desktop' and open the *.m* file in the editor. Below is how the downloaded function works.

```
function [xx, tt] = cosgen2(A, f, phi, t_start, duration, fs)
% COSGEN2 Function to generate Cos wave with varying amplitude, phase
shift
% [xx,tt] = cosgen2(A, f,phi, t_start, dur, fs)
% A = amplitude
% f = desired frequency (Hz)
% phi = phase (in degrees)
% t_start = starting time (in seconds)
% duration = duration of waveform (in seconds)
% fs = sampling frequency (Hz) (Note: number of samples per period = fs/f)
% Returns function value in xx an 'timebase' in tt
if (nargin < 6)
    fs = 50*f;  % gives 50 samples per period by default if fs not specified
end;
```

Notes: The *cosgen2* function takes in phase angle in degrees. The MATLAB '*cos*' function requires phase in radian, so it does a conversion. The first part of the code above allows you to omit the parameter *fs*, which will be set to 50**f* by default.

2. Use the *cosgen2* function to generate the following sinusoid (with start time $t_s = -5\text{x}10^{-3}$, duration = $15\text{x}10^{-3}$ and sampling frequency f_s = 10 kHz)

$$y = 10\cos(1000\pi t)$$

Plot the output 'signal' *y* vs time. Draw this into your lab result sheet.

HINT: MATLAB commands for the above, to be typed in at MATLAB command prompt (>>), are:

[*xx tt*] = *cosgen2*(10, 500, 0, -5e-3, 15e-3 , 1e4);
plot (*tt,xx*)

.

3. Repeat Step 2 with a sampling frequency f_s = 1000 Hz and plot the output. What does the output look like? Why?

4. Repeat Step 2 with
   a. f_s = 800 Hz and
   b. f_s = 500 Hz

   Explain why the output in 4(b) looks the way it does?

B. Fast Fourier Transform (FFT) of sampled signals and windowing effects

1. Connect the output of the function generator directly to the oscilloscope Channel 1 probe.
2. Set the function generator to produce a 50 kHz sinusoid with amplitude 2.0V.
3. Adjust the time base so that the display on the oscilloscope shows exactly **2 cycles** of the waveform.
4. Display the frequency spectrum on the oscilloscope with the following settings:
   i. frequency *Span*: 500 kHz
   ii. *Center* frequency: 250 kHz
   iii. *Scale*: 5dB/
   iv. *Offset*: 0 dBV
   v. *Window*: *Hanning*.

   Note: If you have forgotten how to set these, refer to Lab 1 instruction sheet.
5. Make note of the *FFT Resolution* shown on the display (bottom right corner, just above the softkey menu) and note down in your result sheet.
6. Sketch the displayed spectrum in your lab results sheet.
7. Change the *Window* to *Rectangle*.
8. Sketch the displayed spectrum
9. Explain the shape of the two spectra and any differences based on your post-lab work.

10. Change the time base so that **10 cycles** of the time signal waveform are displayed on the oscilloscope screen. Keep the *Window* function as *Rectangle*.
11. Note the *FFT Resolution* and sketch the spectrum shown in your results sheet.
12. Change the time base so that approximately **50 cycles** of the time signal waveform are displayed on the oscilloscope screen.
13. Note the *FFT Resolution* and sketch the spectrum shown in your results sheet.
14. Discuss the changes to FFT resolution and spectrum shape as the no of cycles displayed is changed.
    Hint: Refer to Module 10 for information on spectral resolution.

15. Change the function generator output to a square wave with 50% duty cycle, maintaining all other settings.
16. Note the *FFT Resolution* and sketch the spectrum shown in your results sheet.
17. Discuss the spectrum shape.

Discussion of experimental results

Based on the results obtained, discuss the effect of sampling and windowing functions on the spectral analysis of signals. What insights has this experiment given you?

~ End of experiment ~